

Using Open Street Map for logistics and planning

A case study

deltamedia
a bpost company

ignotec

Case Study?

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VIENNA 15–17 JULY 2011

- Deltamedia: Delivery company
  - Mostly newspapers



Case Study?

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Case Study?



- 150.000 customers daily
- Independent deliverers
- Not a static situation: changes are daily and significant

Main requirement:

Routing & Planning

Why OSM?



Not sure we could do this!

- Experimentation
- Community

Requirements

- Daily optimisation of individual delivery rounds



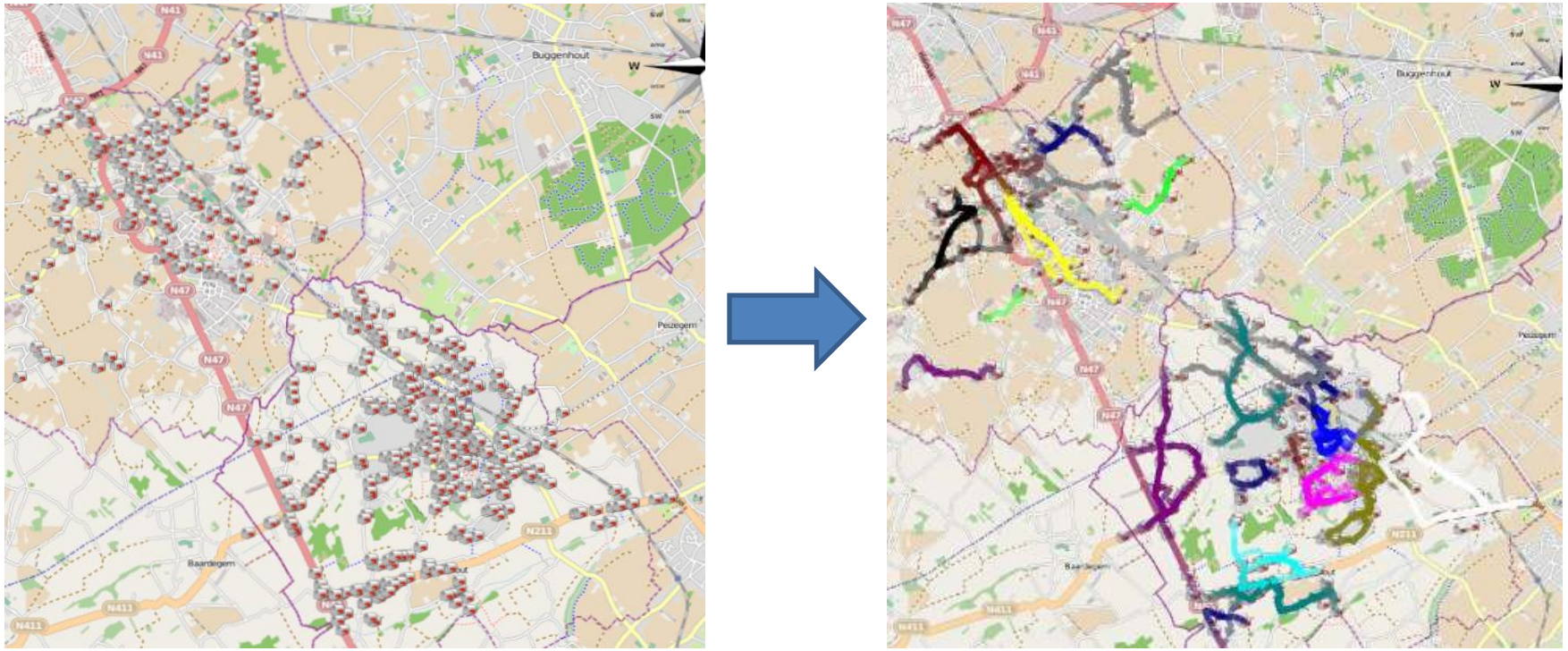
Requirements

- Daily optimisation of individual delivery rounds



Requirements

- New customers/re-optimisation: generate delivery routes given a large number of customers!



Requirements



- Different vehicles types
- Support for construction works
- Clear routing instructions
- Real usable routing results

Automatic optimisation of delivery routes.

Solution

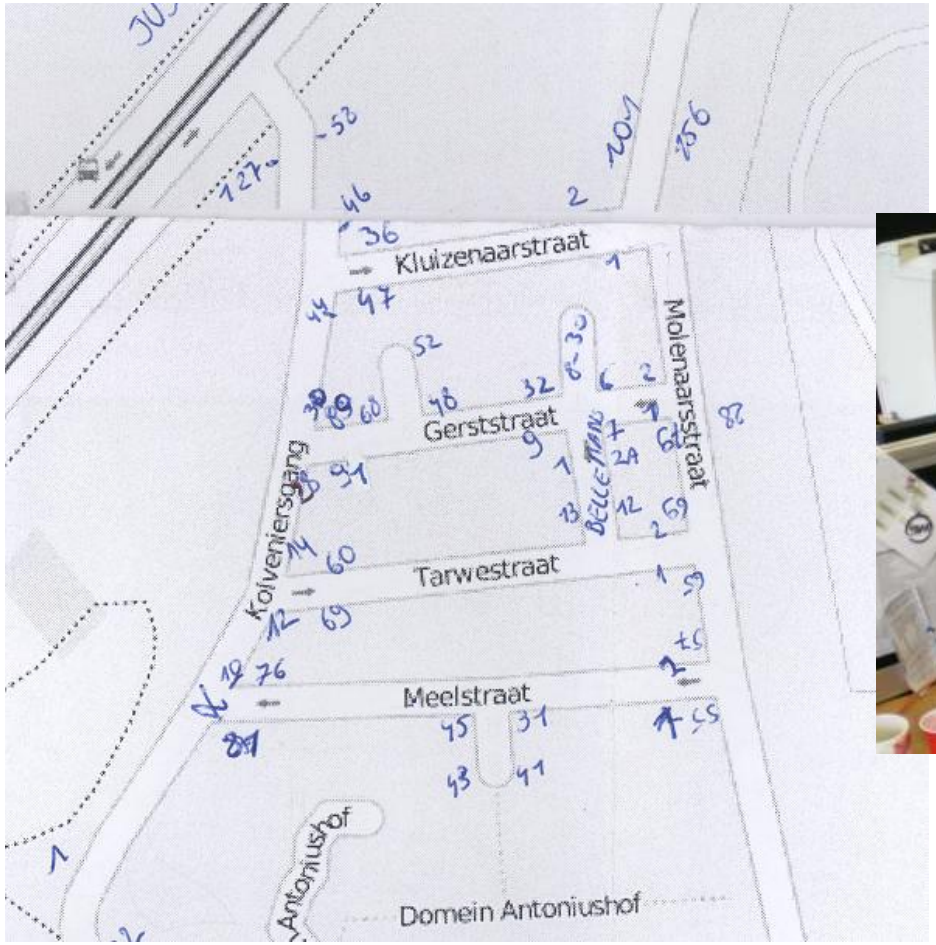


1. Mapping!
2. Routing!
3. Optimisation!

```
// solve the problem!
List<int> solution = solver.Solve(problem);
```

Mapping

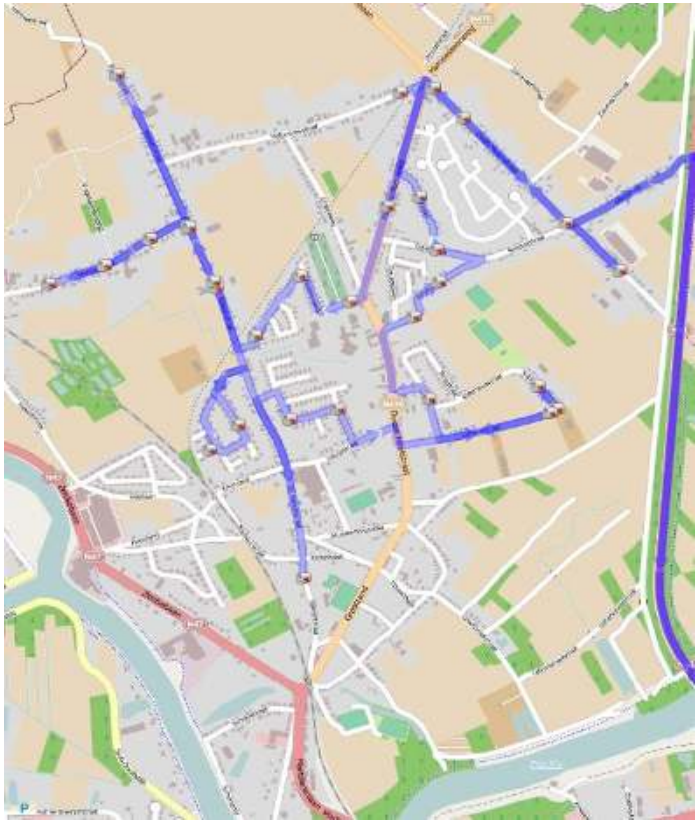
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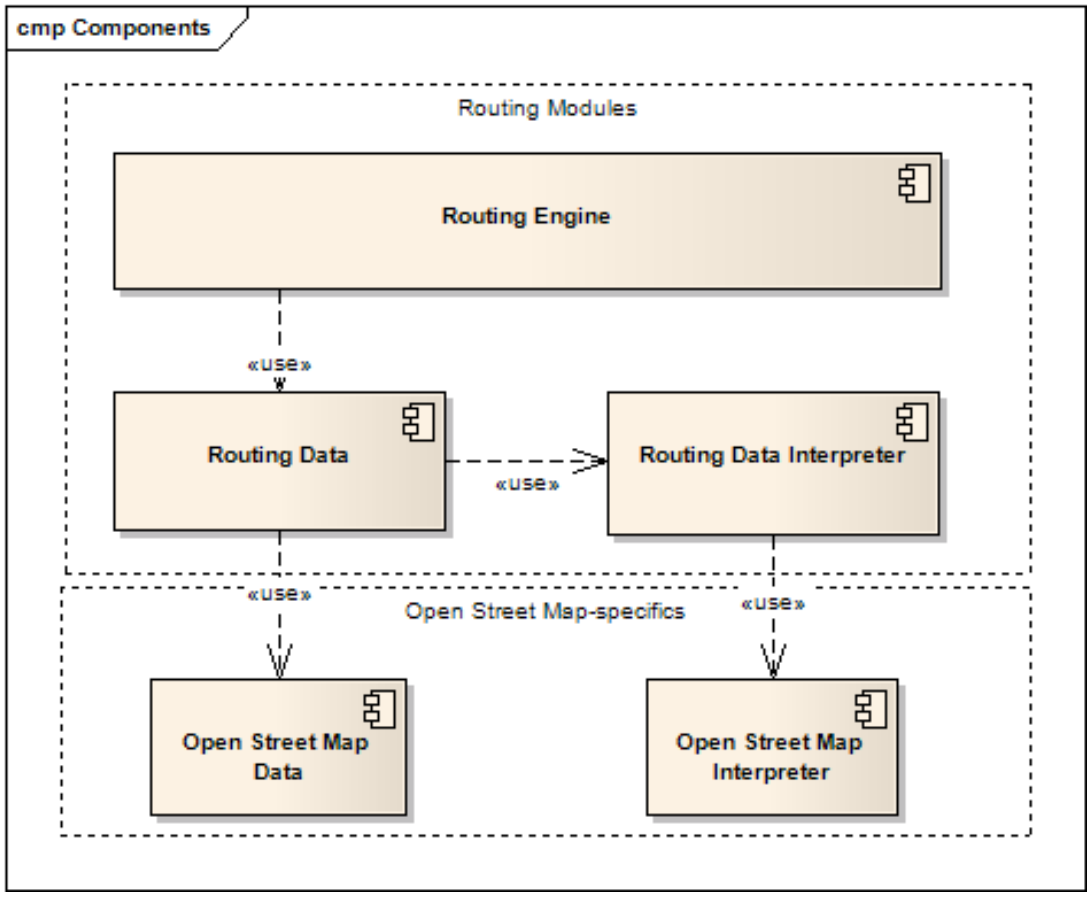


Routing



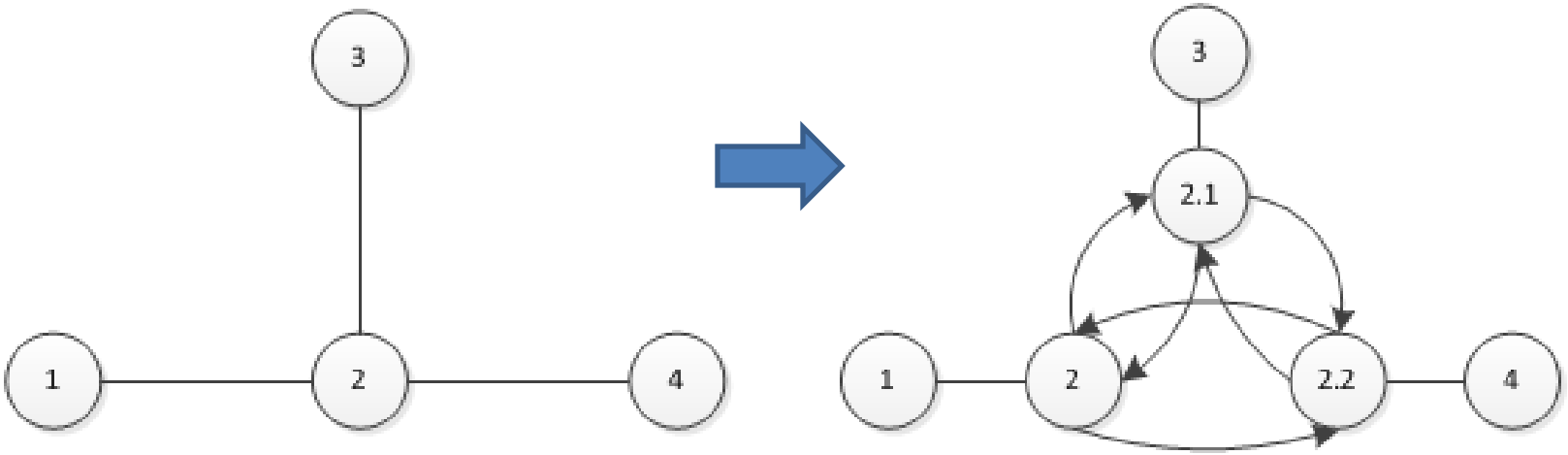


Routing


cmp Components
Routing Modules
Routing Engine
«use»
Routing Data
«use»
Routing Data Interpreter
«use»
Open Street Map-specifics
«use»
Open Street Map Data
Open Street Map Interpreter

Routing

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Optimisation

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Asymetric Travelling Salesman Problem (ATSP)

- *Given the weights between every customer pair, find the shortest route for visiting all customers and returning to the starting point*
- For *n* customers there are *(n-1)!* routes

n	#routes
10	3628800
20	2432902008176640000
91	1.35e140

Optimisation



1. Calculate all weights between all customer pairs using dykstra's shortest path algorithm
   - $\#w \approx n^2$
2. Generate solutions using a best-placement algorithm
3. Optimise using a genetic algorithm

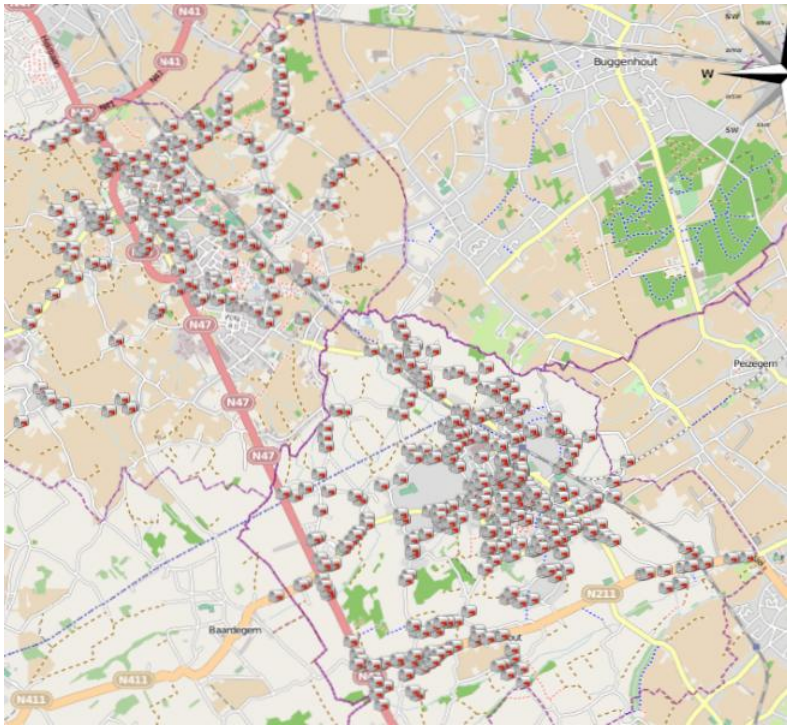
Optimisation

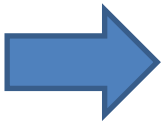
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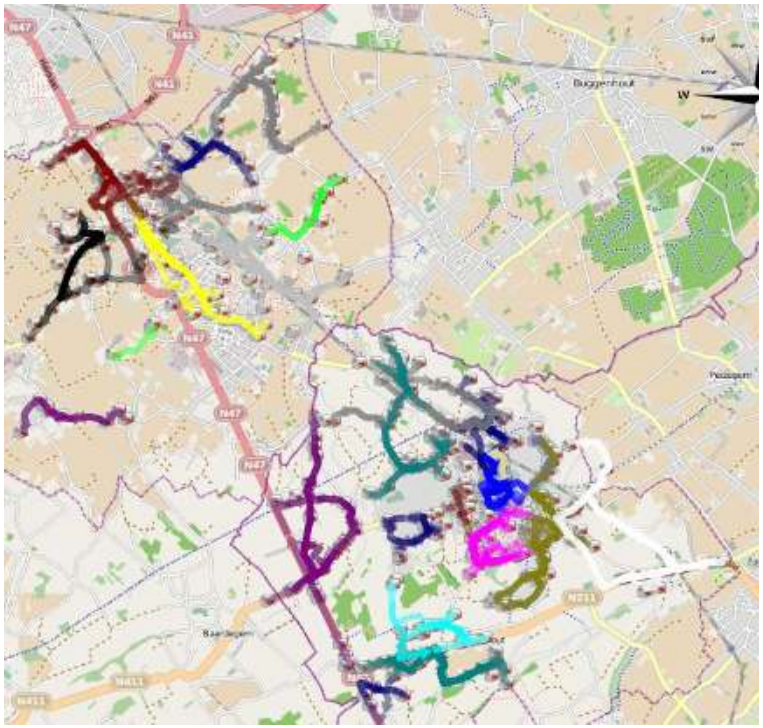
- A Genetic Algorithm: search heuristic that mimics the process of natural evolution
- ‘Evolve’ a population of solutions
  1. Initialisation
  2. Mutation
  3. Reproduction or Crossover
  4. Termination

Optimisation

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Optimisation



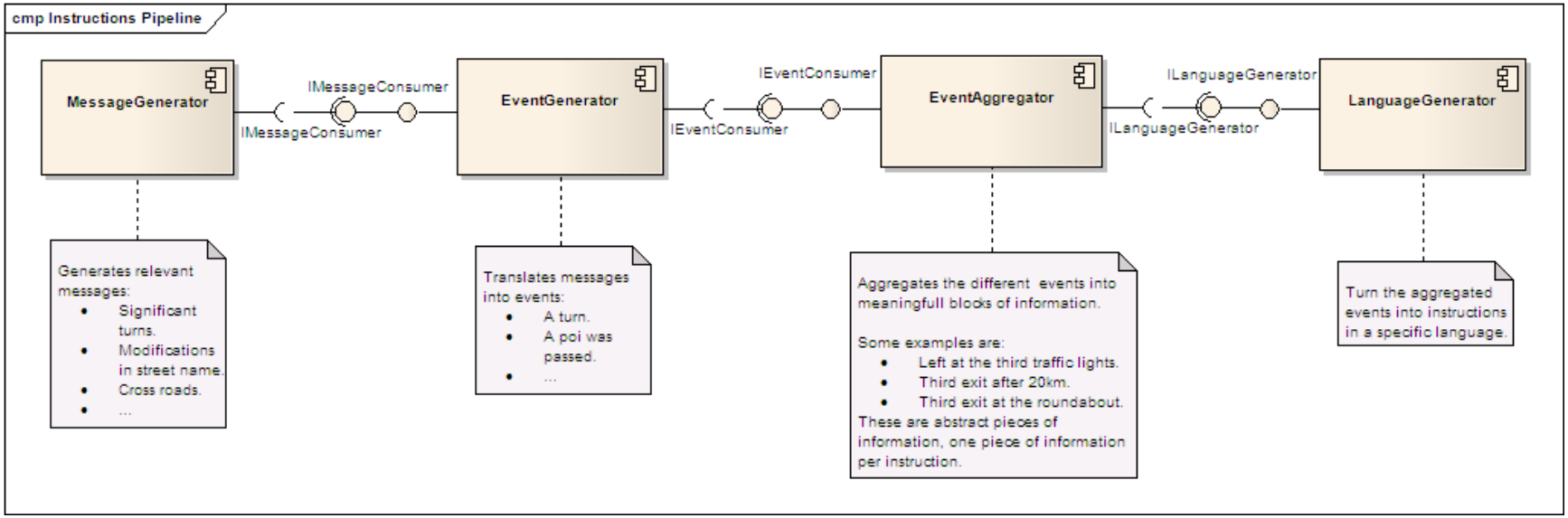
Two cases:

1. Choose a fixed amount of delivery vehicles and divide customers among them.
2. Choose an upper and lower bound on time of one delivery route and calculate the minimum number of needed vehicles.

Instructions

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- A pipelined architecture
- Finite State Machines (FSM)



Quality of solutions

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- Calculations of several scoring metrics per delivery route
  - Distance of address to closest street.
  - Quality of geo-coding result(s).
  - Percentage of unnamed streets.

Future work?

Results



- Fully customized
- Fully integrated
- Library OsmSharp: C#

http://sourceforge.net/projects/osmsharp/



Results



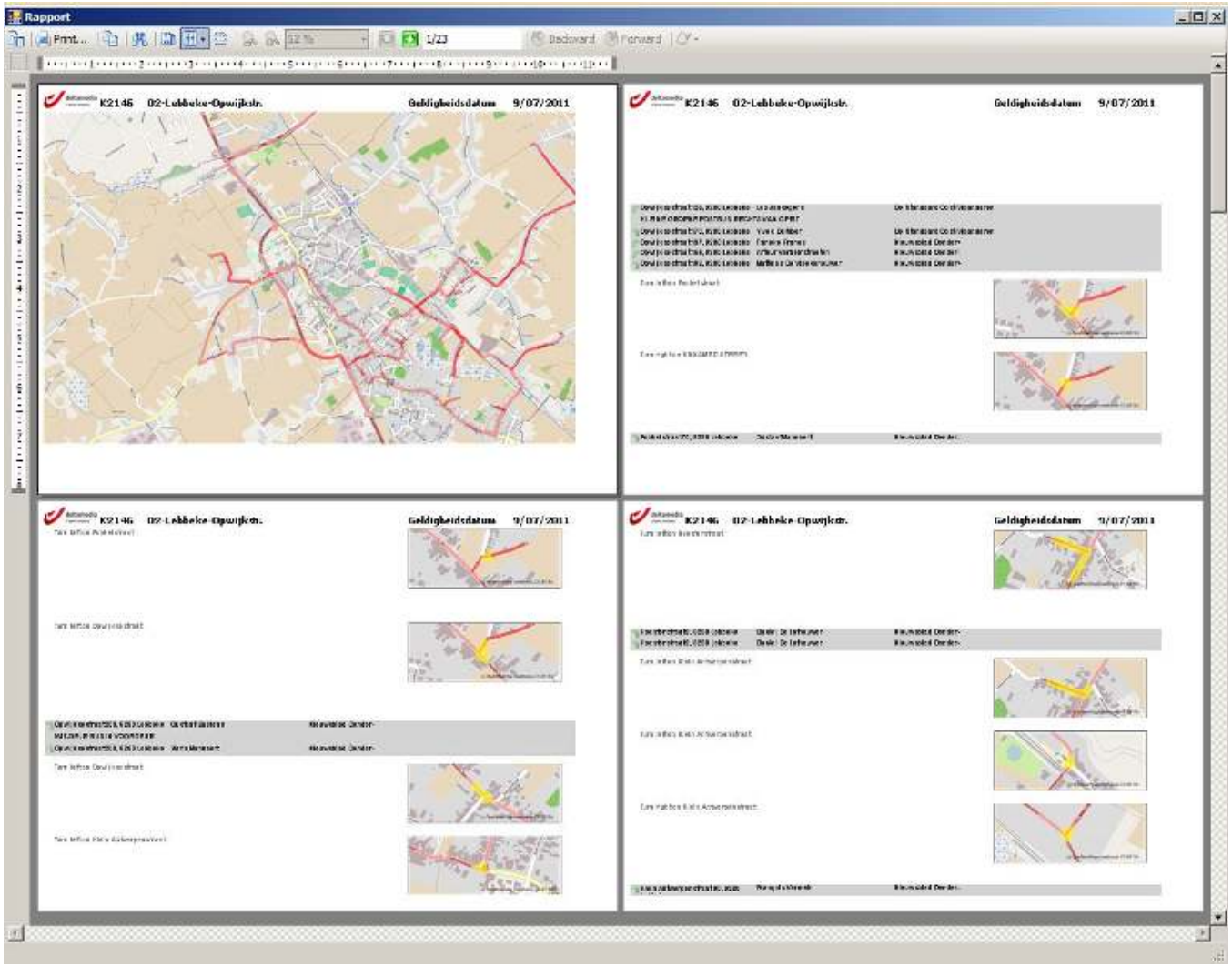
K2115 01-GREMBERGEN-CENTRUM **Geldigheidsdatum: 12/07/11**

HUB: H1 DEPOT: DM202 LEBBEKE ADRES: BRUSSELSESTW 168 BOX 11 9280 LEBBEKE

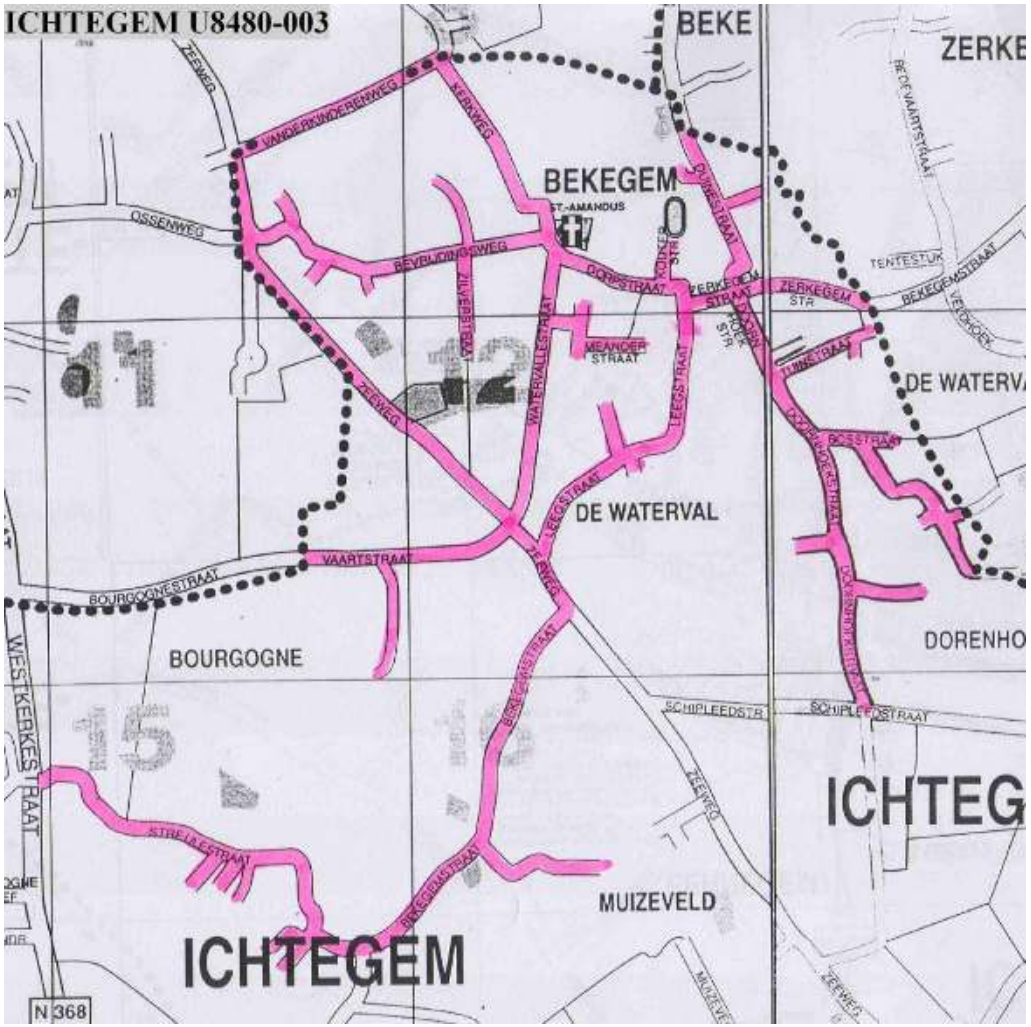
							K2115 Dit rondeschema werd opgesteld door Deltamedia voor intern-administratieve doeleinden. Het werd louter ten titel van inlichtingen ter beschikking gesteld aan de bezorger, die vrij blijft volledig autonoom te beslissen over het door hem te te volgen traject. Er is geen enkele verplichting dit rondeschema te volgen. Elke vorm van reproductie buiten het kader van deze overeenkomst is te strengste verboden.
							START SINT-ELOOISTRAAT
9200	SINT-ELOOISTRAAT	3	R	Nieuwsblad Dender JOBAT	1	123456	VAN CAUWENBERGHE
							1. LINKS GROOTZAND
							1. RECHTS VOORTWEG
9200	VOORTWEG	1B	R	Nieuwsblad Dender JOBAT	1	123456	CLAUS
							TERUG EN RECHTS VERDER GROOTZAND
							VERDER VOLGEN DR HAECKSTRAAT
9200	DOKTER HAEKSTRAAT	51	R	Nieuwsblad Dender JOBAT	1	123456	DE WILDE
							1. RECHTS LOMEERSTRAAT
							EINDE LINKS KOUTERBAAN
9200	KOUTERLAAN	12	R	Nieuwsblad Dender JOBAT	1	123456	BIESEMAN
9200	KOUTERLAAN	13	R	Standaard Oost-Vlaanderen JOBA	1	123456	DE DECKER
9200	KOUTERLAAN	17	R	Nieuwsblad Dender JOBAT	1	123456	QUIDé-VAN OPDORP
9200	KOUTERLAAN	20	R	Standaard Oost-Vlaanderen JOBA	1	123456	BAETENS
							LINKS VOLGEN KLEEMPUTSTRAAT
							TERUG EN 1. LINKS RINGSTRAAT
							1. RECHTS RINGSTRAAT
							TERUG EN 1. RECHTS RINGSTRAAT
9200	RINGSTRAAT	16	R	Nieuwsblad Dender	1	123456	DEPOUILLON
9200	RINGSTRAAT	20	R	Nieuwsblad Dender JOBAT	1	123456	DE KEERSMAECKER
							EINDE RECHTS LOMEERSTRAAT
							EINDE RECHTS DOKTER HAEKSTRAAT
9200	DOKTER HAEKSTRAAT	84	R	Nieuwsblad Dender JOBAT	1	123456	VAN DER MEIRSCH
9200	DOKTER HAEKSTRAAT	72	R	Nieuwsblad Dender JOBAT	1	123456	BAKKERIJ VERSCHEL
							AAN DE KERK RECHTS SMIDSESTRAAT
9200	SMIDSESTRAAT	55	R	Nieuwsblad Dender JOBAT	1	123456	ADVOCATENKANTOO
9200	SMIDSESTRAAT	50A	R	Nieuwsblad Dender	1	123456	VERMEIR
9200	SMIDSESTRAAT	18	R	Nieuwsblad Dender	1	123456	SAEYS
							RECHTS VOLGEN SMIDSESTRAAT
9200	SMIDSESTRAAT	78	R	Nieuwsblad Dender JOBAT	1	123456	MINNEBO
9200	SMIDSESTRAAT	84A2	R	Nieuwsblad Dender JOBAT	1	123456	DE VRIENDT

Datum van afdruk: 12/07/2011

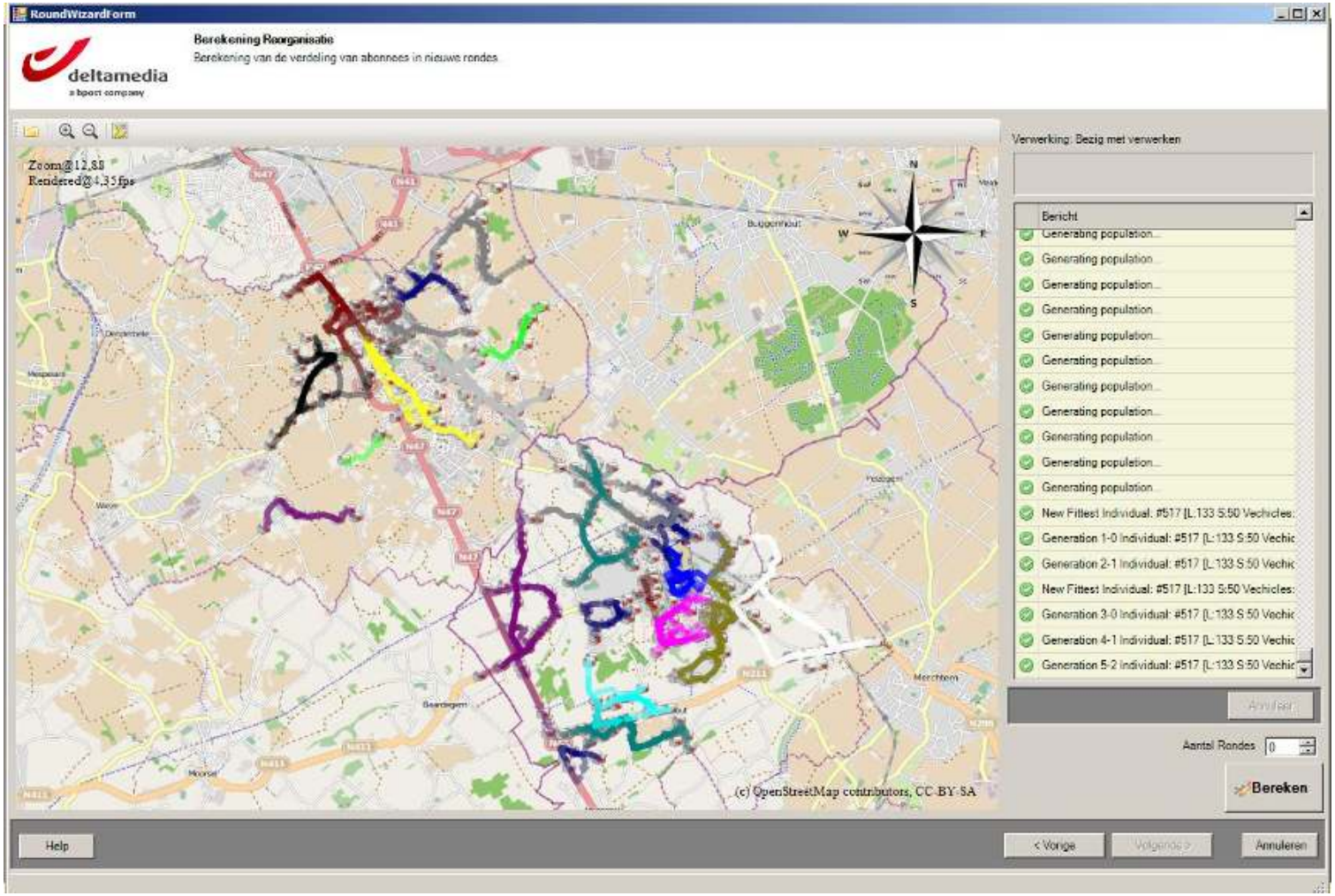
Results


Rapport
K2146 02-Lebbeke-Opwijkstr. Geldigheidsdatum 9/07/2011

Results


ICHTEGEM U8480-003
BEKE
ZERKE
BEKEGEM
ST.-AMANDUS
OSSENWEG
DORPSTRAAT
ZERKEGEM STR
TENTESTUK
BEKEGEMSTRAAT
MEANDER STRAAT
DE WATERVA
DE WATERVAL
LEEGSTRAAT
VAARTSTRAAT
BOURGOGNESTRAAT
BOURGOGNE
WESTKERKESTRAAT
DORENHO
SCHIPLEEDSTR
ICHTEG
MUIZEVELD
ICHTEGEM
N 368

Results


RoundWizardForm
Berekening Reorganisatie
Berekening van de verdeling van abonnees in nieuwe rondes
deltamedia
Zoom@12,88
Rendered@4,35fps
Verwerking: Bezig met verwerken
Bericht
Generating population...
New Fittest Individual: #517 [L:133 S:50 Vechicles:
Generation 1-0 Individual: #517 [L:133 S:50 Vechic
Generation 2-1 Individual: #517 [L:133 S:50 Vechic
Generation 3-0 Individual: #517 [L:133 S:50 Vechic
Generation 4-1 Individual: #517 [L:133 S:50 Vechic
Generation 5-2 individual: #517 [L:133 S:50 Vechic
Aantal Rondes
Bereken
(c) OpenStreetMap contributors, CC-BY-SA
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< Vorige
Volgende >
Annuleren

Future Improvements



- Faster routing
- Adjusted map rendering
- More optimization parameters
- Usability improvements
- Mono
- Better time estimates



Questions?

ben.abelshausen@gmail.com



